

National Statistics Office of Nepal (NSO Nepal)

AI Tools in Official Statistics: Practical Training

Companion worksheet for "Using AI Tools to Prepare Official Statistical Reports"

Trainee Name: _____ Date: _____ Department: _____

Exercise 1: Data Privacy & Ethics (Step 8)

*Instructions: Review the "Would I post this on our website?" test. For each scenario below, decide if it is safe to paste into a free AI tool (like ChatGPT or DeepSeek). Circle **YES** if permitted, or **NO** if prohibited.*

1. You want AI to summarize a 50-page PDF of the already-published *National Population Census 2021* report. **YES NO**
2. You have a spreadsheet containing the names, addresses, and income data of 500 households from an ongoing survey and want AI to look for trends. **YES NO**
3. You are drafting the monthly CPI press release. The data is aggregated, but the official release date is tomorrow. It is strictly embargoed. **YES NO**
4. You wrote a rough draft of a methodology section for a new report. It contains no numbers, just descriptions of your sampling method, and you want AI to improve the grammar. **YES NO**

Exercise 2: Prompt Engineering with RCTF (Step 2)

*Instructions: The following prompt is weak and will likely result in poor AI output. Rewrite it using the **RCTF** framework (Role, Context, Task, Format).*

Weak Prompt:

"Make a paragraph about the foreign trade data. Exports went up 5% and imports went down 2%. It should sound professional."

Your RCTF Prompt:

Role (Who is the AI?): _____

Context (What is the data/situation?):

Task (What exactly do you want?):

Format (Length, tone, specific rules?):

Exercise 3: Data-to-Narrative Drafting (Step 4)

Instructions: You have been assigned to write a short narrative for the Annual Statistical Abstract based on the mock data below. Draft the exact prompt you would type into the AI to generate a 100-word summary.

Data Provided (Mock Tourism Statistics):

Tourism Indicator	2022	2023
Total International Arrivals	614,869	1,014,882
Average Length of Stay	13.0 days	13.2 days
Top Source Country	India (209,105)	India (319,936)

Write your prompt below. (Remember to include the critical instruction: "Use ONLY the statistics I have provided"):

Exercise 4: Fact-Checking AI Output (Step 7)

Instructions: AI tools are writing assistants, not fact sources. Compare the official NSO Nepal mock data against the AI-generated draft below. Identify and list **three factual errors or hallucinations** the AI made.

Official Data (Input):

- Overall Inflation Rate (CPI): 5.4% (year-on-year)
- Food and Beverages Inflation: 6.1%
- Non-food and Services Inflation: 4.9%

AI-Generated Draft (Output):

"In the current fiscal year, Nepal experienced an overall inflation rate of 5.4 percentage points. The data highlights a significant burden on households, driven primarily by the food and beverages sector, which saw a massive inflation increase of 6.5%. Meanwhile, the non-food and services sector remained relatively stable at 4.9%. The rising cost of imported goods was a major factor in this year's economic shifts."

List the 3 errors/hallucinations you found:

1. _____

2. _____

3. _____

Exercise 5: Report Structuring & Outlining (Step 3)

Instructions: You need to start a new chapter on "Agricultural Production" for the upcoming Economic Survey. Write an AI prompt asking it to generate a standard outline. Make sure to specify the NSO context and what the chapter should cover.

Write your prompt below:

Exercise 6: Tone Adjustment & Editing (Step 6)

Instructions: A junior staff member drafted the following paragraph. It has accurate data but uses a very casual, unprofessional tone. Draft a prompt asking the AI to rewrite this paragraph into a formal, official NSO style, while keeping the data exactly the same.

Draft Paragraph (Staff Submission):

"So, the GDP went up a bit this year, like 3.9%, which is way better than last year's 1.9%. The agriculture stuff grew by 2.8% but the services sector did really awesome, growing at 4.6%. This means the economy is bouncing back nicely."

Write your prompt below. (Specify your role, task, and the tone constraints!)

Exercise 7: Drafting a Press Release (Step 5)

*Instructions: You need to issue a press release for the National Population Census. Based on the NSO guide's press release structure, list **five specific formatting instructions** you should include in your prompt to ensure the AI formats the press release correctly for the media.*

1.

2. _____
3. _____
4. _____
5. _____

Golden Rule of AI Assistance

You bring the data. AI brings the words. Always verify every number!

Trainer's Answer Key & Sample Responses

Note for Trainers: For exercises involving prompt drafting (2, 3, 5, 6), trainees' answers will vary. Ensure their responses capture the core intent shown in these samples.

Exercise 1: Data Privacy & Ethics

- **1. YES.** It is already published and public.
- **2. NO.** This is confidential household microdata (PII). Never paste this into AI.
- **3. NO.** The data is aggregated, but it is strictly embargoed. Pasting it to cloud-based AI violates release protocols.
- **4. YES.** There is no data or sensitive information, just methodology descriptions.

Exercise 2: Prompt Engineering with RCTF

- **Role:** You are an official statistician/report writer at NSO Nepal.
- **Context:** Here is recent foreign trade data: Exports increased by 5% and imports decreased by 2%.
- **Task:** Write a short narrative paragraph summarizing this trade data.
- **Format:** Use a formal, professional, official government tone. Keep it under 50 words. Do not add any outside statistics.

Exercise 3: Data-to-Narrative Drafting

"You are an official report writer for NSO Nepal. Here is mock tourism data for the Annual Statistical Abstract: [Trainee indicates pasting the table]. Please write a 100-word formal narrative summary highlighting the total arrivals, the average length of stay, and the top source country.

CRITICAL: Use *ONLY* the statistics I have provided. Do not hallucinate or add any numbers not present in the data."

Exercise 4: Fact-Checking AI Output

1. **Unit Error:** The AI wrote "5.4 percentage points" instead of "5.4%". (A rate is %, a change in a rate is percentage points).
2. **Number Hallucination:** The AI wrote food inflation was "6.5%". The official data says 6.1%.
3. **Fabricated Causation:** The AI invented a reason for the shift ("rising cost of imported goods"). This was not in the data provided. AI must not guess economic drivers.

Exercise 5: Report Structuring & Outlining

"You are an official report writer at NSO Nepal. I am preparing a new chapter on 'Agricultural Production' for the upcoming Economic Survey. Please generate a complete section-by-section outline for this chapter. For each section, provide a heading, 2-3 bullet points on what to cover, and a suggested word count. The structure should match standard NSO government publications."

Exercise 6: Tone Adjustment & Editing

*"You are an expert editor for NSO Nepal government publications. Please rewrite the following draft paragraph to make it highly formal, professional, and suitable for an official statistical report. **CRITICAL:** Keep all the statistics exactly the same. Do not add any new facts or data. Just improve the tone and vocabulary."*

Exercise 7: Drafting a Press Release

- **1.** Include a strong, single-sentence headline with the most important finding.
- **2.** Open with a lead paragraph stating who, what, when, and the key figure.
- **3.** List key findings as 3-5 short bullet points or paragraphs.
- **4.** Include a brief methodology note (e.g., sample size, collection period).
- **5.** End with official contact information for NSO Nepal.